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LETTER TO THE EDITOR

Retiring M1/M2: a view from the succinate receptor

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To the Editor

Chen and colleagues report that tumour-derived succinate, acting through its receptor SUCNR1, drives M2 polarization of tumour-associated macrophages and thereby confers cetuximab resistance in colorectal cancer [1]. The experiments are careful and the causal chain is plausible. I write to question one link the conclusion depends on entirely: the designation of the responding macrophages as "M2." The M1/M2 classification is an extreme oversimplification of macrophage biology, and the succinate–SUCNR1 axis the authors place at the centre of their mechanism is one of the clearest demonstrations of why.

The M1/M2 distinction came from mouse bone-marrow macrophages exposed to single stimuli in culture. Outside that setup, macrophages do not split into two camps. They occupy a continuous activation space, co-express "M1" and "M2" markers, and switch phenotype with their surroundings. The same marker is regularly made by cells doing opposite jobs, which is why an M1/M2 label often predicts very little about function. Single-cell and spatial work has largely replaced the binary with a continuum [2].

Succinate and its receptor SUCNR1 (GPR91) make the point starkly. Consider four tissues. In adipose, SUCNR1 activation calms inflammation and supports type-2 responses [3]. In skin, removing GPR91 changes inflammatory disease in opposite directions across models, which is hard to reconcile with succinate being a uniformly pro-inflammatory signal [4]. In tumours, including the colorectal setting Chen et al. examine, the axis does push macrophages toward a pro-tumoural state, consistent with their findings and with earlier work [5]. In the kidney, however, the same axis drives fibrosis through a profibrotic macrophage programme and CTGF-dependent crosstalk with fibroblasts, producing pathological matrix deposition rather than repair [6]. One ligand, one receptor, opposite outcomes across tissues. The "M2" state that fits the tumour case cannot be the same biological state that, elsewhere, resolves inflammation in fat or lays down a fibrotic scar in kidney.

The binary breaks more deeply still. In fibrotic disease, macrophages do not merely shift phenotype; they cross cell-type boundaries entirely, transdifferentiating into myofibroblasts through macrophage-to-myofibroblast transition (MMT), a process now dissected at the single-cell and lineage-tracing level and identified as a major driver of pathological remodelling [7]. No M1/M2 label can accommodate a cell that is no longer a macrophage.

The reason the framework cannot hold is that the determinant of outcome sits upstream of any polarity label, in metabolism itself. Succinate is part of the succinate dehydrogenase and itaconate node that controls the IL-1β–HIF-1α response, and that node operates at the border between inflammation and repair [8]. Which way it tips is set by context. Itaconate, the metabolite next door, is anti-inflammatory in cultured bone-marrow macrophages and pro-inflammatory in lung-resident ones [9]. Same molecule, opposite direction, decided by where the cell happens to be. If a metabolite can do that, no pre-assigned M1/M2 label is going to hold.

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A practical alternative

What would work better has been in the literature for a decade and just needs to be used. Murray and colleagues argued in 2014 that macrophages should be described by what activates them, in which tissue, and with which markers, rather than placed on a binary axis. In the present case, the responding cell would be identified as a succinate-activated macrophage in the colorectal tumour microenvironment, with the receptor (SUCNR1) and a defined marker panel specified alongside [10]. Single-cell and spatial methods now let one go further and map the response onto a measured continuum, with no need to choose between bins that were never empirically clean [2]. And because the direction of the response is set by metabolism rather than by surface phenotype, the metabolic node carrying the signal, here the succinate dehydrogenase and itaconate axis, belongs in the cell's description rather than appearing as a footnote to an M1/M2 label.

This is not perfectionism; it is what Chen et al.'s own therapeutic proposal requires. They suggest SUCNR1 as a target to overcome cetuximab resistance. Yet the same receptor calms inflammation in adipose tissue and drives fibrosis in the kidney, so a SUCNR1-directed therapy will help or harm depending entirely on tissue context, and the M1/M2 framing is precisely what makes that context disappear from view. Naming the state by its stimulus, its tissue, and its markers, rather than by a pole, is what turns their finding into a safely targetable one. This strengthens their case rather than weakening it.

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Declarations

Competing interests The authors declare no competing interests.

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